HRISHIN DEBNATH

Student Researcher | Multilingual AI and Language Models

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PROFILE

Higher-secondary science student in Dhaka. I am building a Bangla version of a medical exam question set to test whether language models answer worse in Bangla than in English. I am also the sole author of two papers on Dhaka's air quality, both under review. I am looking for a professor to guide my next study on language disparity in LLMs and on image-generation models.

Research interests: language disparity in LLMs (Bangla and other low-resource languages against English), LLM evaluation, machine translation quality, image-generation models

Next study: go deeper into language disparity in LLMs, building on the translated MedQA set, and extend it to image-generation models: does a model produce worse images from a Bangla prompt than from the same prompt in English?

RESEARCH

Catching Translation Errors in Bangla Medical Exam Questions (working title) **Jul 2026 – Present**

Researcher, mentored study at CreateBud Inc. In preparation

- Building a Bangla version of the 12,000-question MedQA exam set; in a 500-question pilot of 5 open models, the revised prompt cut flagged translations for the chosen model from 399 to 100.
- Tested the error checks with planted mistakes: they caught 50 of 50 swapped answer options and 36 of 50 tenfold number changes.

Next-Day PM2.5 Forecasting in Dhaka, Bangladesh, from a Single Reference-Grade Monitor: A Provenance-Audited Gradient-Boosting Approach Using ERA5 Reanalysis and Open Satellite Fire Data **Sep 2026**

Sole author. Submitted to Air Quality, Atmosphere & Health (under review) | **Code**

- Beat the persistence baseline by 5.22 µg/m³ RMSE (30.80 vs 36.02, R^2 0.795) on a 12-month holdout with XGBoost, using 248,953 hourly rows merged from OpenAQ, ERA5, NASA POWER and NASA FIRMS.
- Checked the result with walk-forward testing and an independent LightGBM model tuned with Optuna.

Does Dhaka's Air Get Cleaner When the City Empties? A Nine-Year Natural Experiment Using Eid Holidays and Reference-Grade PM2.5 Data **Sep 2026**

Sole author. Columbia Junior Science Journal (under review) | **Code and data**

- Found PM2.5 falls 36% on average when Dhaka empties for Eid (95% CI 22% to 47%), across 16 holidays and 75,325 hourly U.S. Embassy readings, using an event study and 2,000 placebo runs.

RESEARCH EXPERIENCE

CreateBud Inc. **Remote, part-time**

Research Intern *Jul 2026 – Present*

- Running the Bangla medical translation study (see Research) with a CreateBud mentor.
- Tested CreateBud's pre-release co-working platform and reported 6+ bugs in its project, team and task features.

TECHNICAL SKILLS

Programming: Python (pandas, NumPy, SciPy, scikit-learn, statsmodels), R, SQL, DAX; working knowledge of JavaScript and TypeScript

ML & statistics: XGBoost, LightGBM, Random Forest, time-series forecasting, walk-forward validation, Optuna, hypothesis testing, event studies

Data & tools: PostgreSQL, MySQL, MongoDB, Power BI, Tableau, Excel, Power Query, Power Automate, Git, Jupyter

SELECTED PROJECTS

Desk Watch: on-device study-focus app (in development) **Electron, TypeScript, MediaPipe**

- Windows app that uses the webcam, fully on the device, to sound an alarm when the user picks up a phone, leaves the desk or looks away; includes a site blocker, planner and Bangla interface.
- 76 commits and 219 unit tests. Co-written with Claude Code: I wrote about 60% of the code, set the requirements, made the design decisions and tested each stage.

NASA Space Apps Challenge 2026: Baseline (Team Obsidian Prime, in progress) **Python, JavaScript**

- Technical lead on a prototype that compares an astronaut's daily health readings with their own normal range and cites the NASA study behind every flag.

OTHER EXPERIENCE

VibeWeave (early-stage startup) **Remote, part-time**
Market Analyst *Jun 2026 – Present*

- Built market studies of India's textile industry for five regions (fabric prices, raw-material commodities, listed companies) and a verified database of 26 clothing manufacturers.

Elevvo Pathways | **Work on GitHub** **Remote, part-time**
Data Analyst Intern *Aug 2025 – Sep 2025*

- Segmented 1,998 retail customers with RFM analysis in Python; 306 "Champions" generated 23.81% of revenue.

CORViS (student-led non-profit) | **corvisbd.org** **Dhaka**
Deputy Director, Research & Development *Dec 2024 – Present*

- Organised 5+ seminars in the Speaks Academia series by inviting professors and industry professionals, including a panel on careers and AI in economics.

EDUCATION

St. Joseph Higher Secondary College, Dhaka **2026 – 2028 (expected)**
Higher Secondary Certificate (HSC), Science, English version

Cosmo School, Dhaka **2026**
Secondary School Certificate (SSC), Science, English version. GPA 5.00 out of 5.00 (1,160 of 1,250 marks)

WorldQuant University | **Credential** **Dec 2025 – Jun 2026**
Applied Data Science Lab

- Completed eight end-to-end projects, each passed at 90% or higher: regression, classification, clustering, time-series and GARCH forecasting, A/B testing, and serving models through an API.

CERTIFICATIONS

Certificates: Google Data Analytics (Oct 2024), **Google Business Intelligence** (Nov 2024), **freeCodeCamp Data Analysis with Python** (Jan 2024), **freeCodeCamp Relational Database** (Jul 2024), **freeCodeCamp JavaScript Algorithms and Data Structures** (Sep 2024), **HackerRank SQL (Advanced)** (May 2024), **Cisco Data Analytics Essentials** (Apr 2024), **LinkedIn Advanced Power BI: DAX** (Sep 2024)

Job simulations (Forage): Quantium Data Analytics (Aug 2025), **Tata Data Visualisation** (Jul 2024), **Accenture Data Analytics and Visualization** (Jun 2024)